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Fundamentals of Radio Communications

112-BOLCEW08

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Concrete Experience

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TLO – Terminal Learning Outcome



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The student will define Radio Communications fundamentals in a classroom environment, given a networked computer workstation and appropriate references by demonstrating understanding of basic components of Radio Communications, Modulation, and Selective Squelch, Communications Security and common COTs and Military radios in order to support EW effects while achieving a minimum of 70% on all Practical Exercises (PE) and end of module exam.

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Learning Step Activities



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- Define the Basic Components of Radio Communications
- Define Radio Communication Principles
- Define Modulation
- Define Selective Squelch
- Define Basic COMSEC Principles
- Identify Commercial Off the Shelf (COTS) Technologies
- Identify Common Military radios

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Basic Concepts of Radio Communications



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Radio Communications Defined



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Radio Communications are:

- Nonvisual Information that is transmitted across space using electromagnetic waves
- A Near instant form of communication (on earth)
- A cornerstone and necessity of modern society
- Essential to military and government operations

Key Layers of radio communication :

- Radio (Transmitter, Receiver and Transceiver)
- Radio Link
- Channel
- Network



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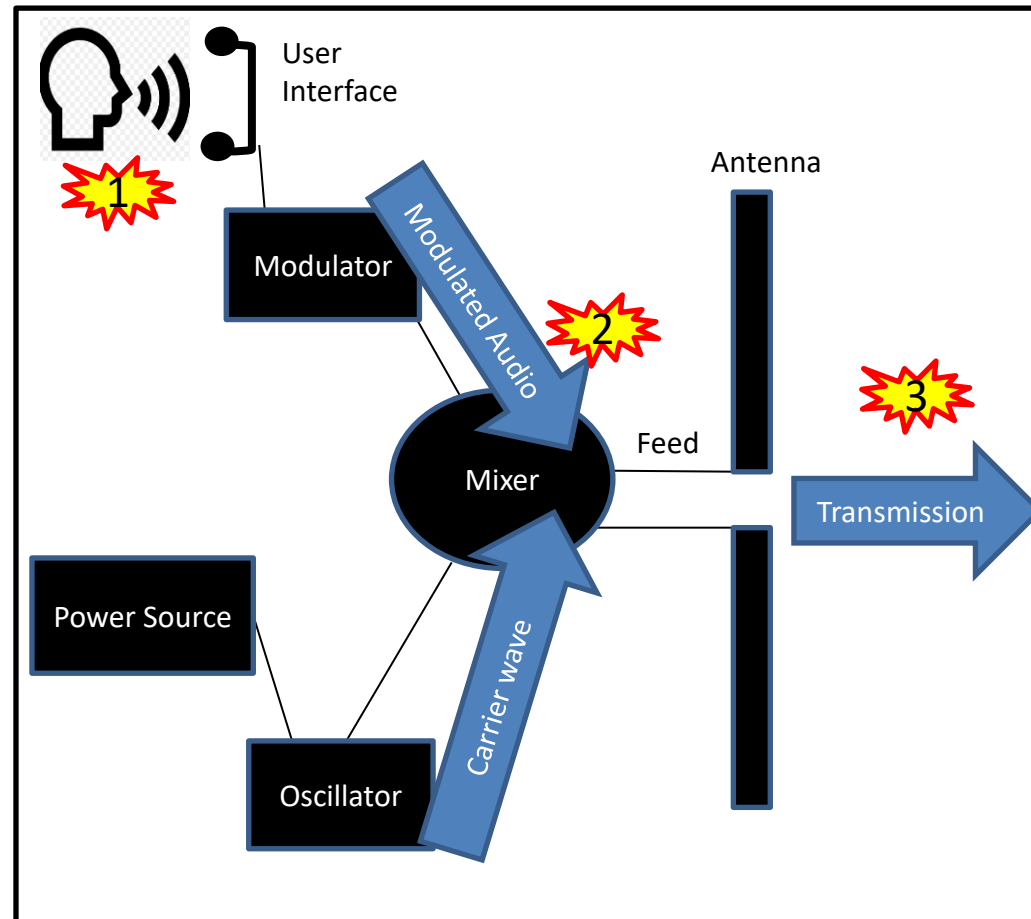
A Radio is:

- A device that transmits or receives nonvisual information through electromagnetic waves
- Three basic types
 - Transmitter (only transmits)
 - Receiver (only receives)
 - Transceiver (both)

Basic Components:

- Power Source
- Antenna
- Feed Line
- Modulator/Demodulator
- Wave Generator (Oscillator)
- Mixer
- User interface i.e.(microphone)

Basic Radio Diagram



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Radio Link



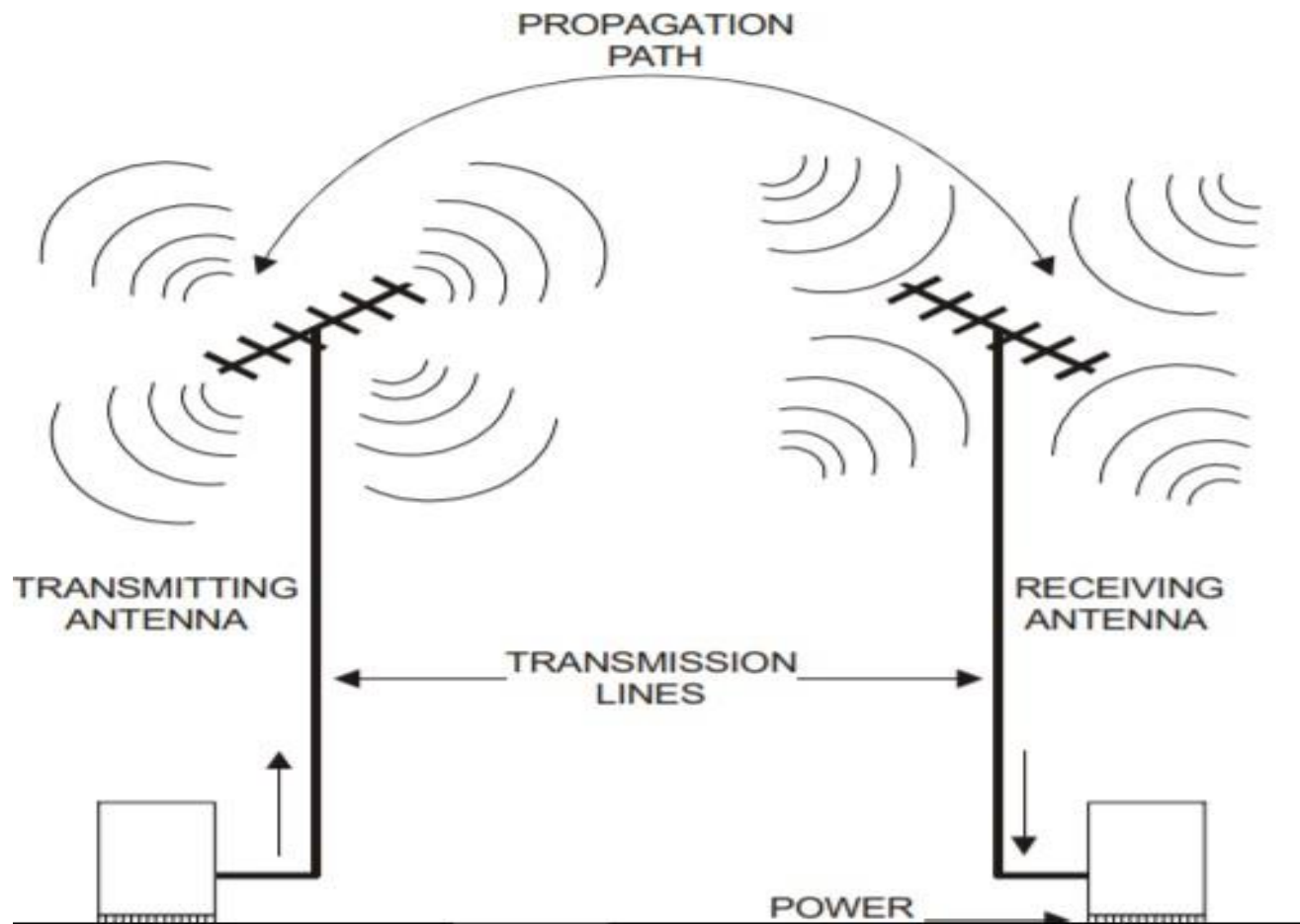
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A Radio Link is:

- A wireless circuit (connection) formed by at least two radios

Components of a Link:

- Transmitter
- Transmission Lines
- Transmitting Antenna
- Propagation Path
- Receiving Antenna
- Receiver



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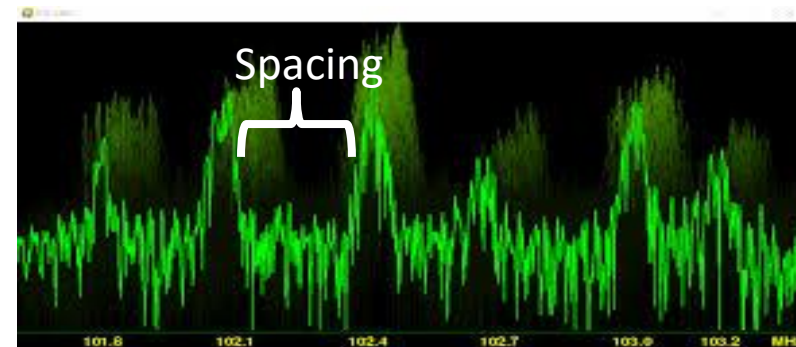
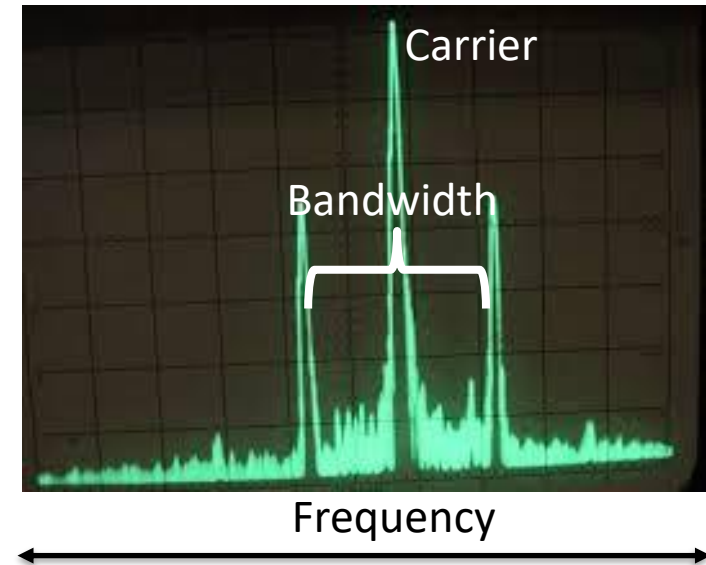
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A Channel is:

- A bandwidth of frequency over which a radio link occupies
- A channel can be composed of two frequencies depending on duplexing (more on that later)

Key Terms:

- Bandwidth:
(range of frequency a channel occupies)
- Carrier Frequency:
(Center unmodulated frequency)
- Modulation:
(How the carrier frequency is changed)
- Channel Spacing:
(Bandwidth between two channels)



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Questions?

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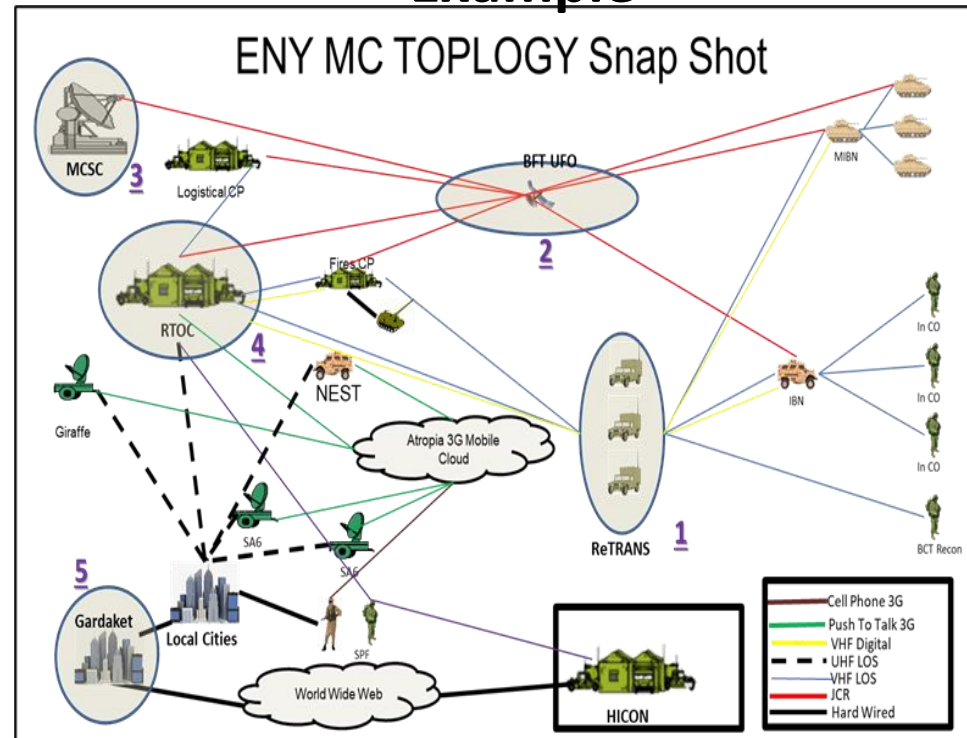
Radio Network:

- A collection of radios linked by one or more radio channels
- Can be very simple or very complex
- Can use one or multiple channels at a time.
- Can allow for multiple users at a time.
- Can be composed of multiple sub networks.

BLUF:

- Radio networks are more art than science and can vary substantially in size and purpose!

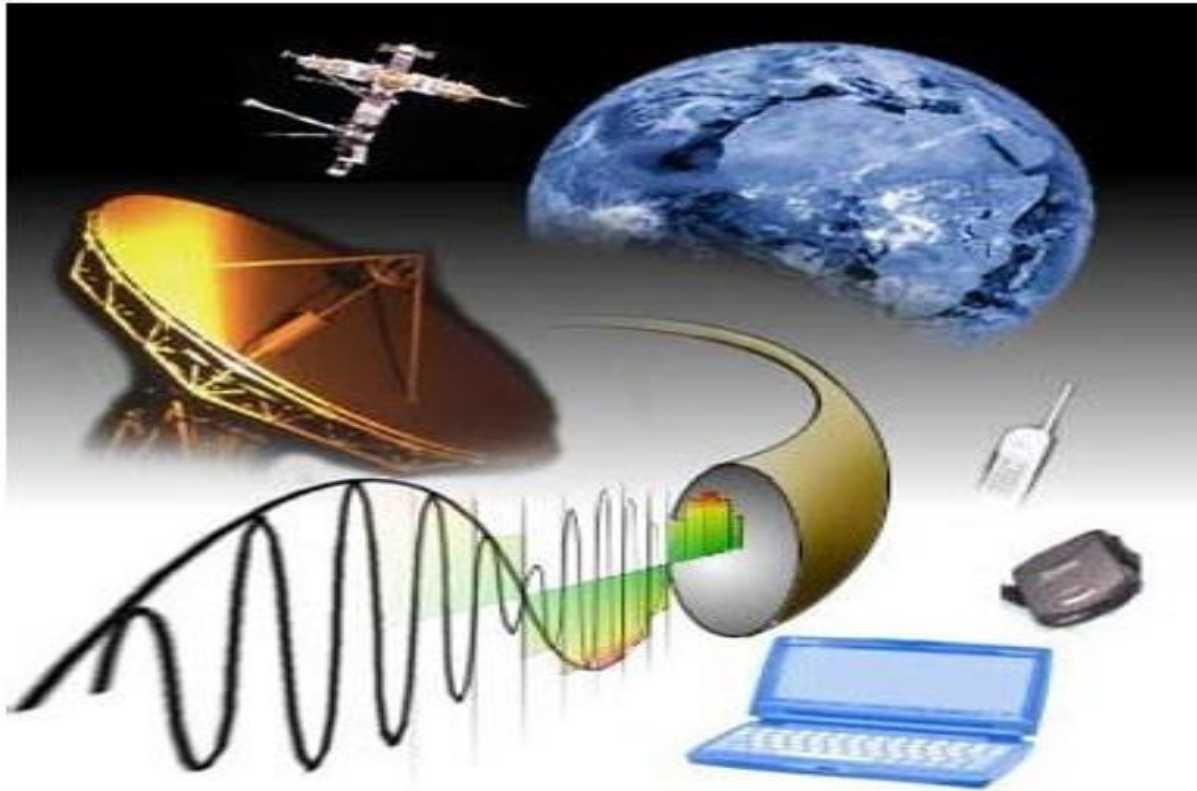
Example



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Expanded Concepts of Radio Communications

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Duplexing



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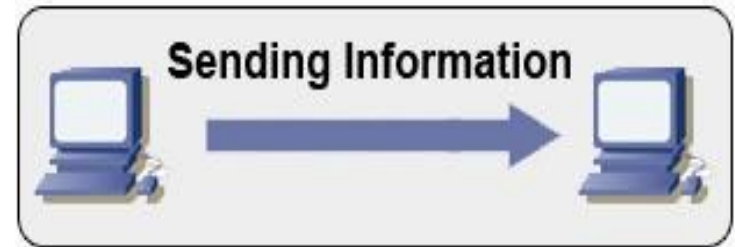
Duplexing:

- The ability for Radios in a network to communicate back and forth.
- If radios can communicate only one way (Broadcast), the network is called Simplex

Duplex has two modes:

- Half Duplex
- Full Duplex

Half-Duplex



Full-Duplex



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SIMPLEX (Broadcast) Mode:

- Allows only one station to transmit in one direction **always**
- These systems are often employed for:
 - TV
 - Commercial Radio
 - Mass Messaging



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Half-DUPLEX

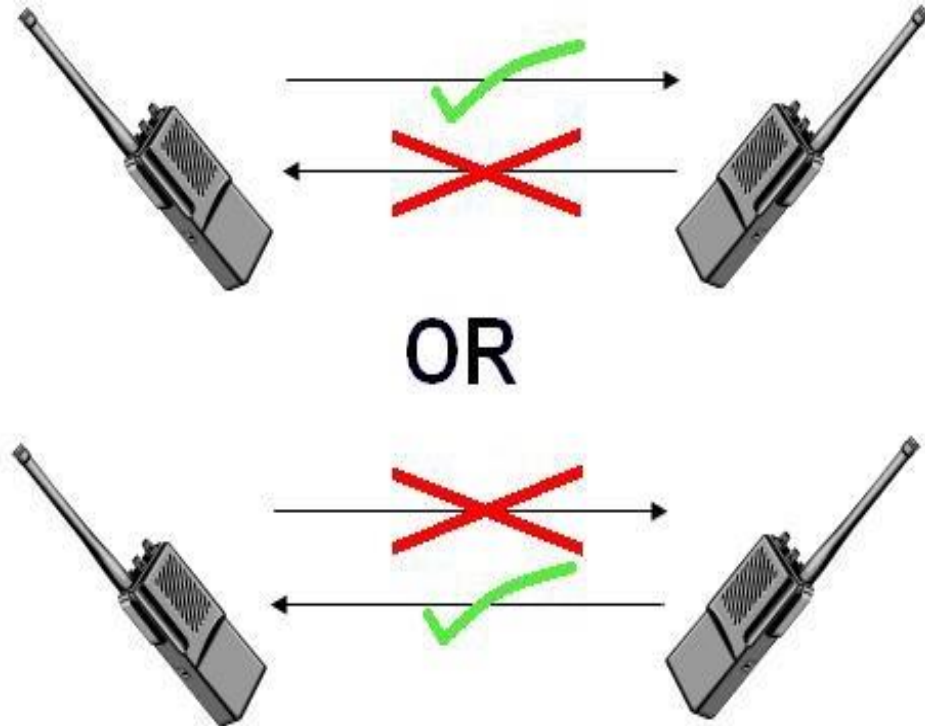


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Half- DUPLEX (HDX) is:

- Type of communication that allows communications in both directions **but:**
- Only one at a time
- Not simultaneously

Both stations are transmitting on the same frequency (channel) each time.



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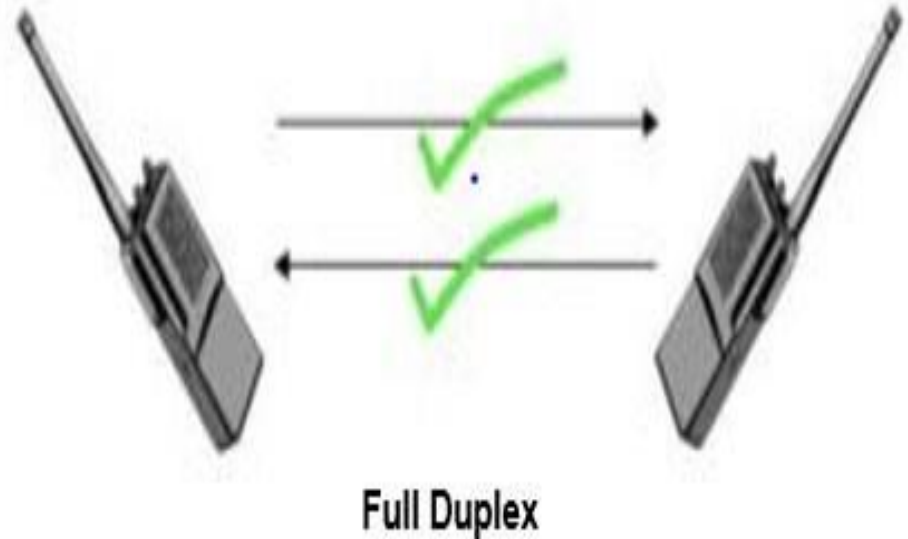
Full DUPLEX



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Full-DUPLEX (FDX):

- Allows communications in both directions, both at the same time or simultaneously
- Uses TWO different frequencies for Transmit and Receive
- Two way radios are great examples, same as phones (land-line and cellular)



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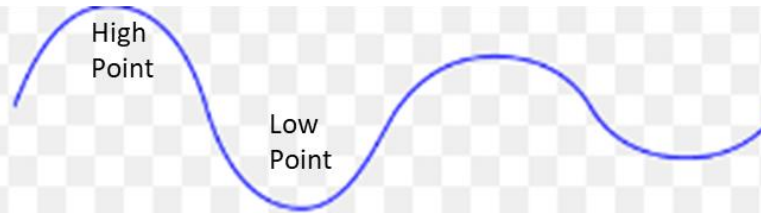
Modulation



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Modulation: The Process of changing a constant frequency carrier wave according to changing input wave by **mixing** the two signals.

Modulating Wave



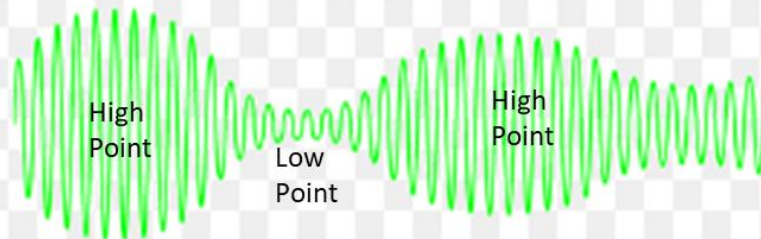
Analog Input (Voice)

Carrier Wave



Constant Frequency

Amplitude Modulation



Changing Amplitude

Frequency Modulation



Changing Frequency

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Mixing



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Heterodyne (mixing) is:

- The process combining a low frequency wave (audio signal) with a high frequency wave (carrier) in order to produce a resultant higher frequency wave
- Accomplished by mixing original modulated carrier with a sine wave of another frequency
- Can also be done in reverse

History:

- In 1901, Reginald Fessenden demonstrated the first ever heterodyne receiver to making Continuous Wave signals audible. His theory was that it creates new frequencies by combining or MIXING two frequencies

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Modulation and Mixing Expanded



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A Carrier Wave is:

- An unmodulated wave that can be modified to represent transmitted information:
- The frequency for a radio signal is the carrier wave's center frequency

Input Wave:

- A low frequency wave that is encoded with information (generally analog)
- The input wave generally goes through a “step up” before mixing in order to increase its frequency

Modulated Wave:

- A resultant high frequency wave after the Input wave has been modified by a modulator
- Modulated waves can contain digital or analog information
- Modulated waves are mixed with the carrier waves in a mixer before being transmitted

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Analog and Digital Information

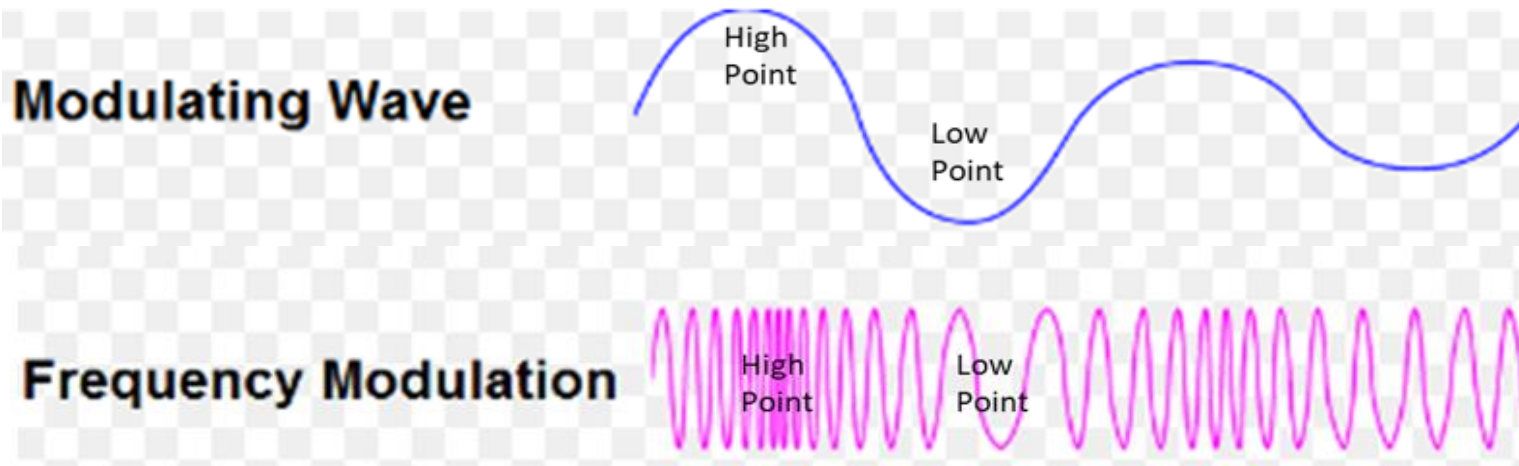


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Analog Information: A **Continuous** measurement of a changing input signal in which the information **directly correlates** to changes from the input.

Digital Information: A **sampled measurement** of a changing input signal in which the information **is measured and quantized** in discrete states i.e. (1, 0) to represent changes from the input.

Analog Example



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Modulation Types



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There are 3 major types of modulation types each with different sub-techniques

Amplitude Modulation (AM)

- Double-sideband modulation with unsuppressed carrier (used on the radio AM band)
- Double-sideband suppressed-carrier transmission (DSB-SC)
- Double-sideband reduced carrier transmission (DSB-RC)
- Single-sideband modulation (SSB, or SSB-AM), very similar to single-sideband suppressed carrier modulation (SSB-SC)
- Amplitude Shift Keying (ASK)

Frequency Modulation (FM)

- Analog FM
- Frequency Shift Keying (FSK)

Phase Modulation (PM)

- Analog Phase Modulation
- Phase Key Shifting

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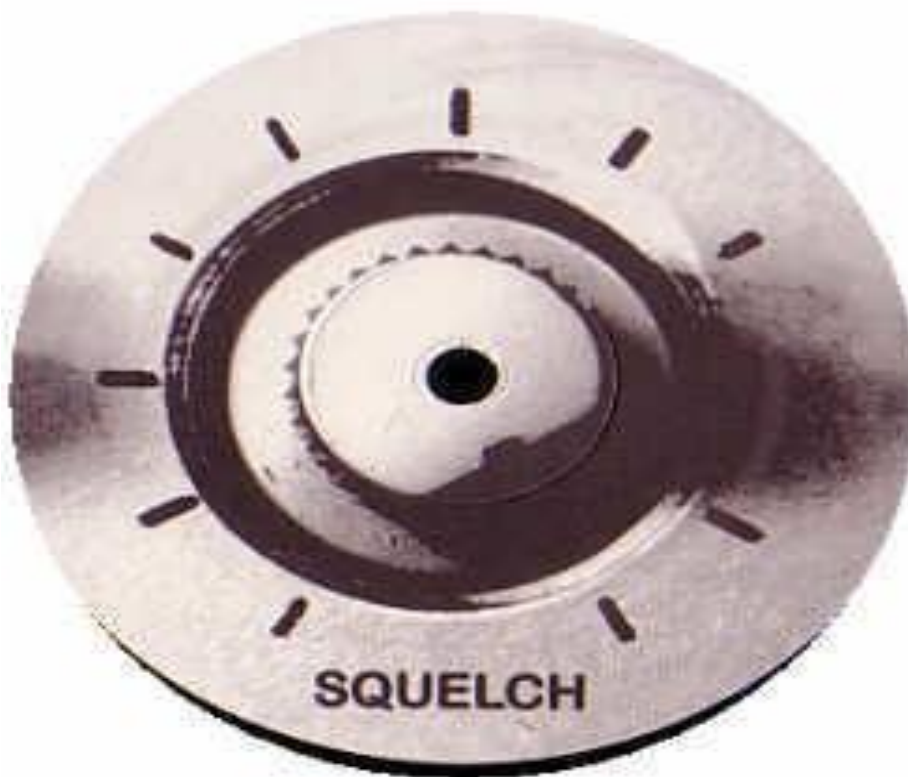
Questions?



Limiting Interference

Selective Squelch

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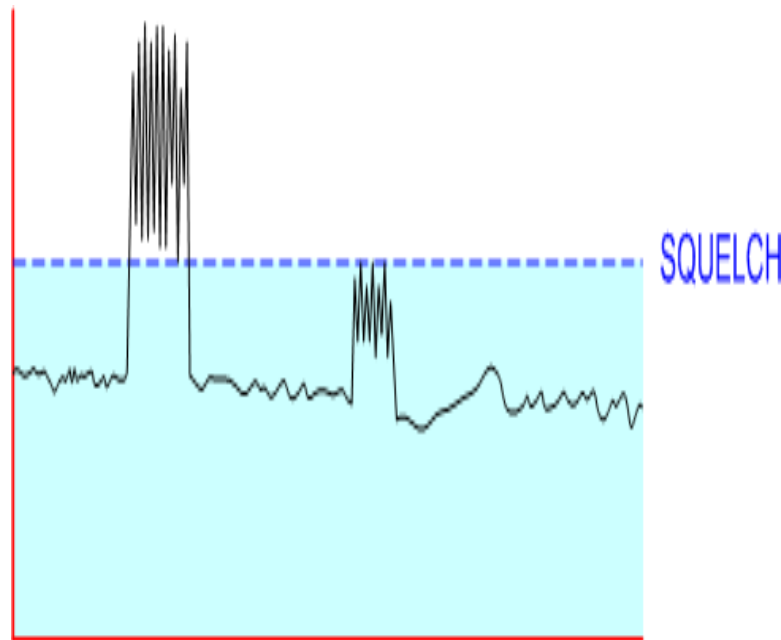
Squelch



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Definition of SQUELCH:

- A circuit that cuts off the output of a receiver when there is no input or is lower than a specified limit
- **Squelch is a noise gate** that only allows signals at a specified strength over a threshold to be played through the speaker



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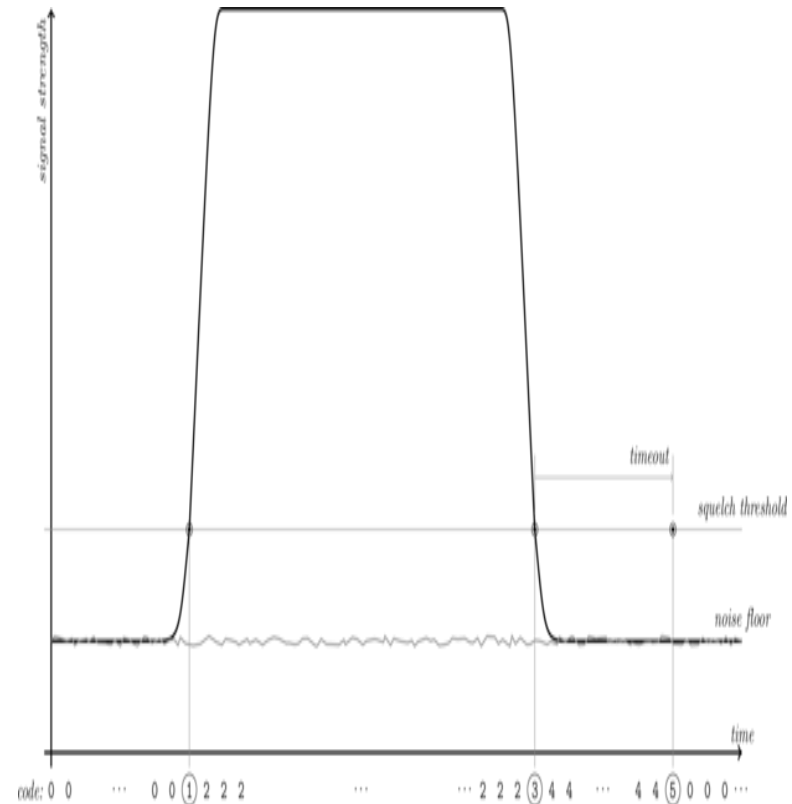
Squelch



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A simple carrier or noise squelch:

- Can operate strictly on the signal strength of the signal such as when a walkie-talkie mutes the audio when no signal is present
- Be pretest or manual
- Most often, the squelch threshold is set with an adjustable knob marked squelch



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Tone Squelch and Selective Calling



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Tone squelch, or other forms of selective calling, are sometimes used to solve interference problems.

Five different techniques are commonly used:

- **Continuous Tone-Coded Squelch System (CTCSS)**
- Selective Calling (SELCALL)
- Digital-coded Squelch (DCS)
- Expanded Tone-Coded Squelch System (XTCSS)
- **Dual Tone Multi-frequency (DTMF)**

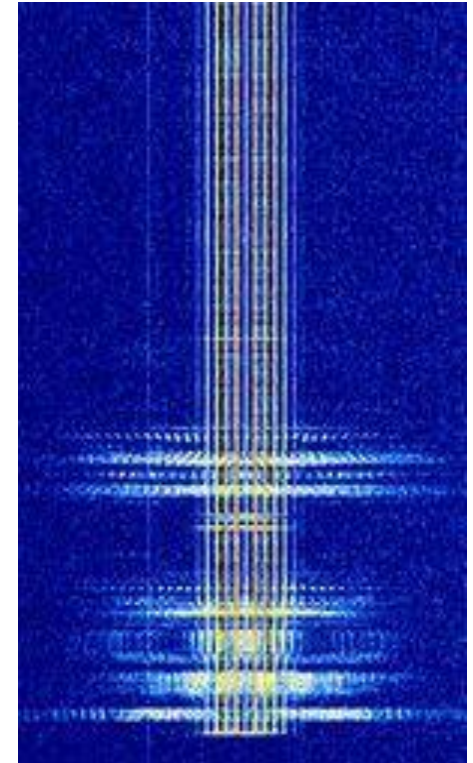
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Continuous Tone-Coded Squelch System



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- Continuous Tone-Coded Squelch System (CTCSS) is the most common squelch control technology
- For multiuser channels:
 - CTCSS filters out other users (receivers) if they are using a different CTCSS Tone or no CTCSS
 - Supper imposes any 50 low-pitch audio tones on the transmitted signal, ranging from 67 to 257 Hz
 - The tones used may be referred to as sub-audible tones



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CTCSS Tones



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Commonly used tones (in HZ):

67.0	88.5	110.9	141.3	179.9	225.7
71.9	91.5	114.8	146.2	186.2	229.7
74.4	94.8	118.8	151.4	192.8	233.6
77.0	97.4	123.0	156.7	203.5	241.8
79.7	100.0	127.3	162.2	206.5	250.3
82.5	103.5	131.8	167.9	210.7	256.3
85.4	107.2	136.5	173.8	218.1	

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Dual Tone Multi-frequency (DTMF)



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- Radios with DTMF decoders may monitor all system traffic or remain muted until called
 - Depending on the system design
 - Radio receives the correct digit string, it may momentarily buzz
- Some systems use DTMF for push-to-talk unit ID
 - When push-to-talk is pressed, the radio sends a string of DTMF digits
 - Radio has a unique string of digits
 - Allows the base station to know who last called/pressed the push-to-talk
- DTMF selective calling:
 - Radio is alerted by a string of digits
 - Uses 2- to 7- digits
 - Can be dialed from a traditional telephone dial connected to a radio

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DTMF Keypad Frequencies



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The DTMF Keypad is laid out in a 4×4 matrix

Each row represents a low frequency; each column represents a high frequency

These tones are then decoded by the switching center to determine which key was pressed

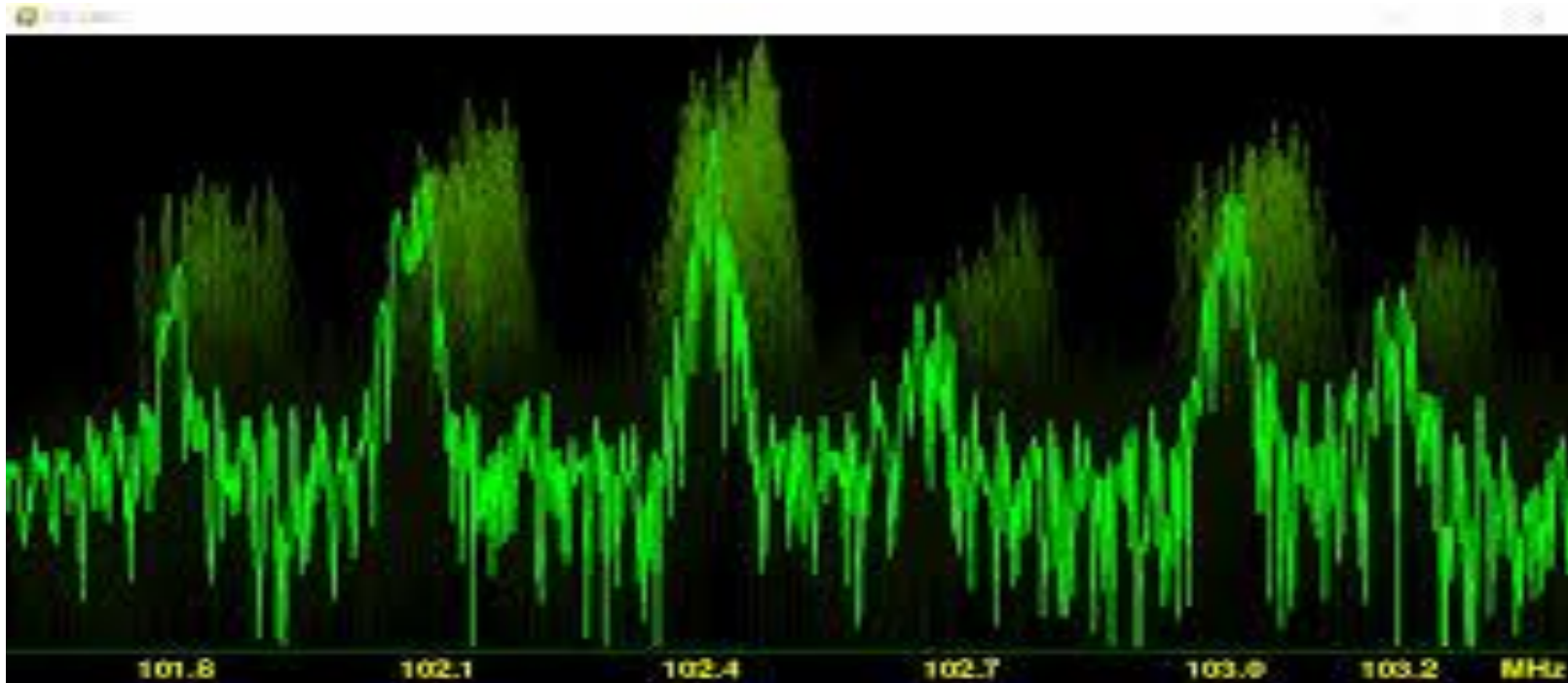
	1209 Hz	1336 Hz	1477 Hz	1633 Hz
697 Hz	 1	 2	 3	 A
770 Hz	 4	 5	 6	 B
852 Hz	 7	 8	 9	 C
941 Hz	 *	 0	 #	 D

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Communications Security



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Communications Security



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Communications Security (COMSEC): A technique of encrypting radio communications so signals can only be properly demodulated through use of a shared key or chipper

- All military radios should use COMSEC
- Many Civil agencies and Civilian radios use COMSEC
 - Digital Mobile Radio (DMR)
 - TETRA
 - Professional Mobile Radio 25 (PMR25)
- COMSEC can contain multiple Keys and Ciphers

BLUF:

- COMSEC protocols significantly increase the security of your transmission
- No radio or security technique is 100% effective (but pretty close)

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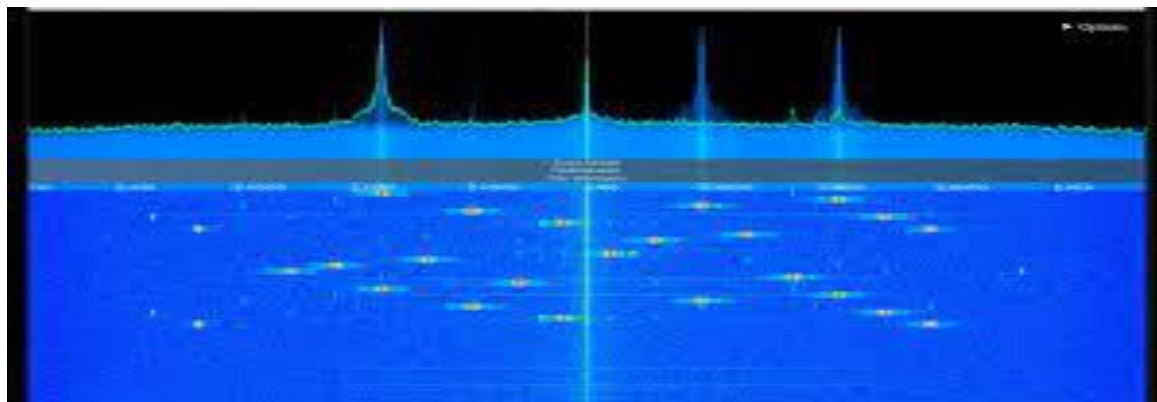
Frequency Hopping



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Frequency Hopping (Frequency Agile): A technique in which radios can seamlessly and continuously maintain a radio link by moving through a number of predesignated frequency's during each transmission

- Can be very slow (a few frequencies per second)
- Can be very fast (thousands of frequencies per second)
- Can follow a set and predictable pattern
- Can be Sudo random
- Requires some sort of shared protocol and precise timing between networked radios (COMSEC)



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Frequency Hopping



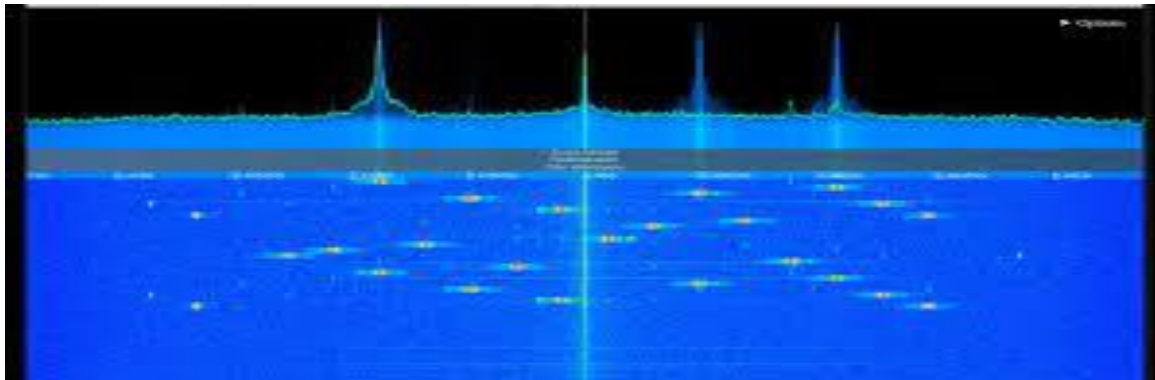
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Applications:

- Frequency Hopping significantly reduces interference on radio
- Makes a radio hardened (but not invulnerable) to Jamming
- Significantly reduces probability of intercept (ability to identify/listen in)
- Harder to geo locate through direction finding

BLUF:

- Frequency Hopping is a necessity for most military radios and should always be used when available
- A Combination of both Frequency Hopping and COMSEC provides military radios with multiple layers of security



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Commercial Off-the-Shelf (COTS) Technologies



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Commercial Off-the-Shelf (COTS)



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COTS:

- Any item of supply (including construction material) that is
 - A commercial item
 - Sold in substantial quantities in the commercial marketplace
 - Offered to the Government

Radio Communications:

- Military, paramilitary, and other adversaries are increasing use of civilian commercial-off-the-shelf (COTS) technology
- A basic understanding of these systems and the technology that enables them is vital to EW mission success

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Networked Radio Systems



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The Networked Radio Systems have three (3) sub categories:

- Family Radio Service (FRS)
- General Mobile Radio Service (GMRS)
- Professional Mobile Radio (PMR)



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Family Radio Service (FRS)



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FRS defined as:

- Private two-way, short-distance voice and data communications
- Facilitating family or group activities
- Does not need a Gov't license to operate
- Widely available



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General Mobile Service (GMRS)



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General Mobile Radio Service (GMRS):

- Land-mobile UHF Radio service available for short-distance two-way communications
- FCC expanded GMRS to allow short data messaging apps including text messaging and GPS location information



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Professional Mobile Radio (PMR)



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PMR:

- Label given to mobile radio networks that are used by a single organization or entity
- Field radio systems that uses portable, mobile, base station, and dispatch console radios
- Operates on VHF or UHF frequency bands
- Typical examples are the radio systems used by policemen and firemen

PMRs data:

- 2.0 watt power/range of up to 25 miles
- Uses both FRS and GMRS frequencies
- 22 channels with 121 privacy codes for 2,622 combinations
- 10 call tones PTT interruption feature
- Point to MULTI-POINT!

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Common Military Radios



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AN-PRC-150



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Common Name: HF radio

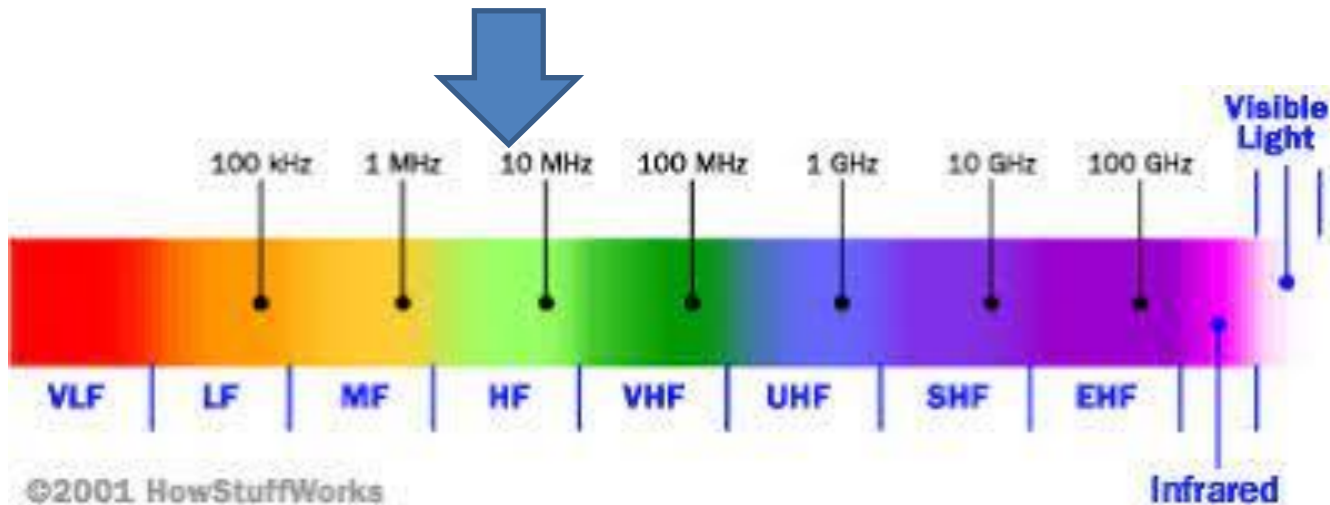
Frequency Range: 3-30 MHz

Primary Use: Beyond LOS tactical Voice and Data communication

Common Lowest Echelon: Company

Modulation: AM, ASK

Max Power: 20W



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AN-PRC-1523E



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Common Name: FM/ASIP

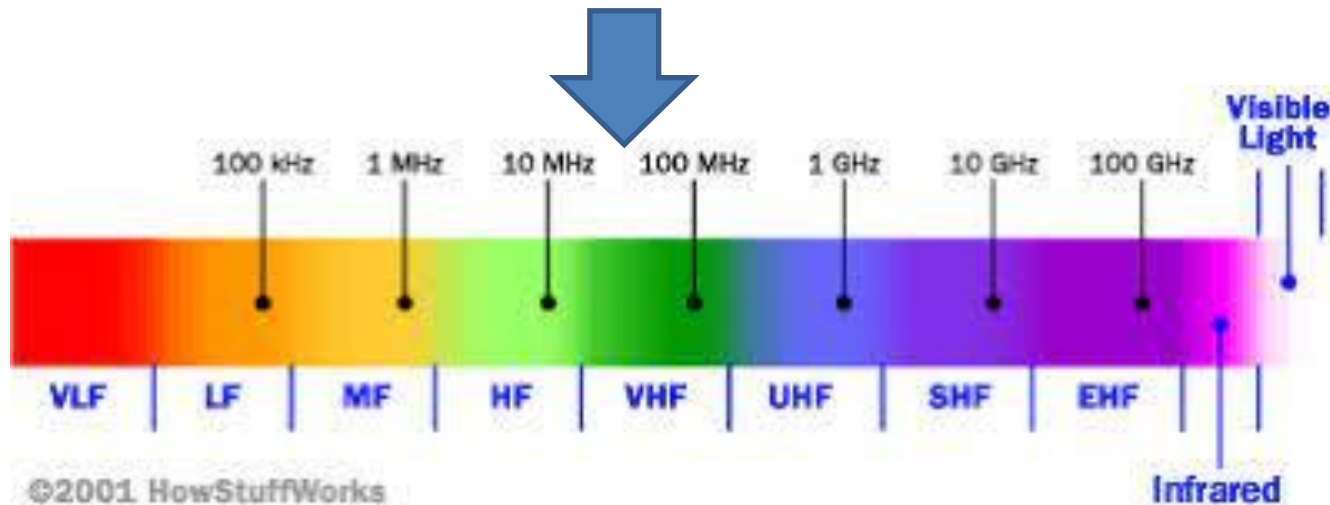
Frequency Range: 30-88 MHz

Primary Use: LOS tactical Voice and data communication

Common Lowest Echelon: Platoon

Modulation: FM, FSK

Max Power: 50W



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AN-PRC-117G



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Common Name: TACSAT

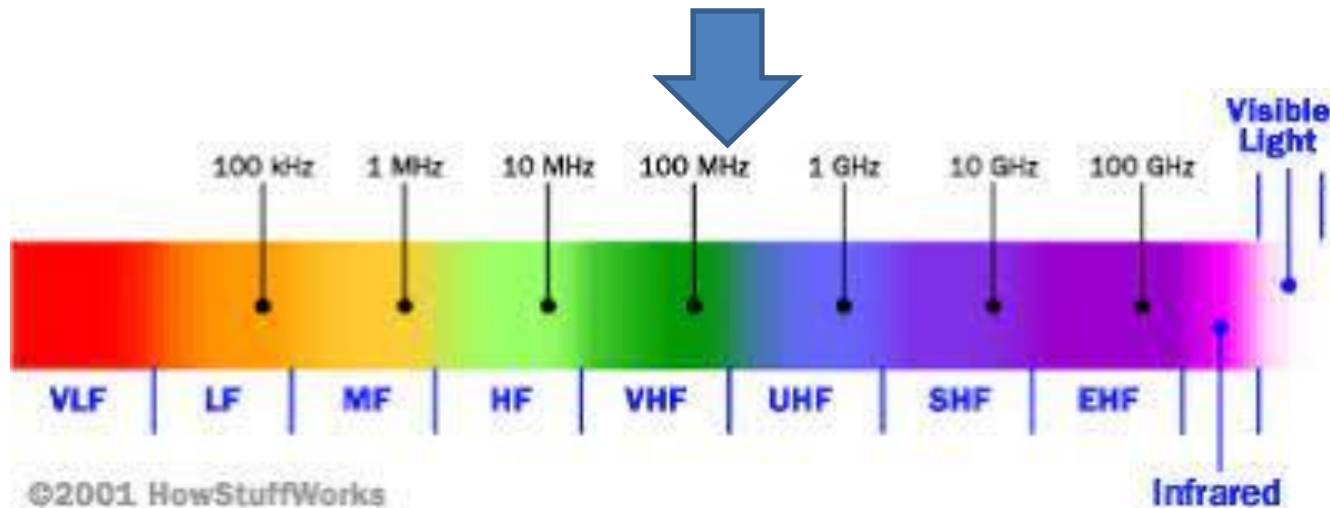
Frequency Range: ~200-300

Primary Use: LOS tactical Voice and data communication

Common Lowest Echelon: Company

Modulation: FM, FSK

Max Power: 10W (for UHF)



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Joint Battle Command Platform



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Common Name: JBCP/BFT

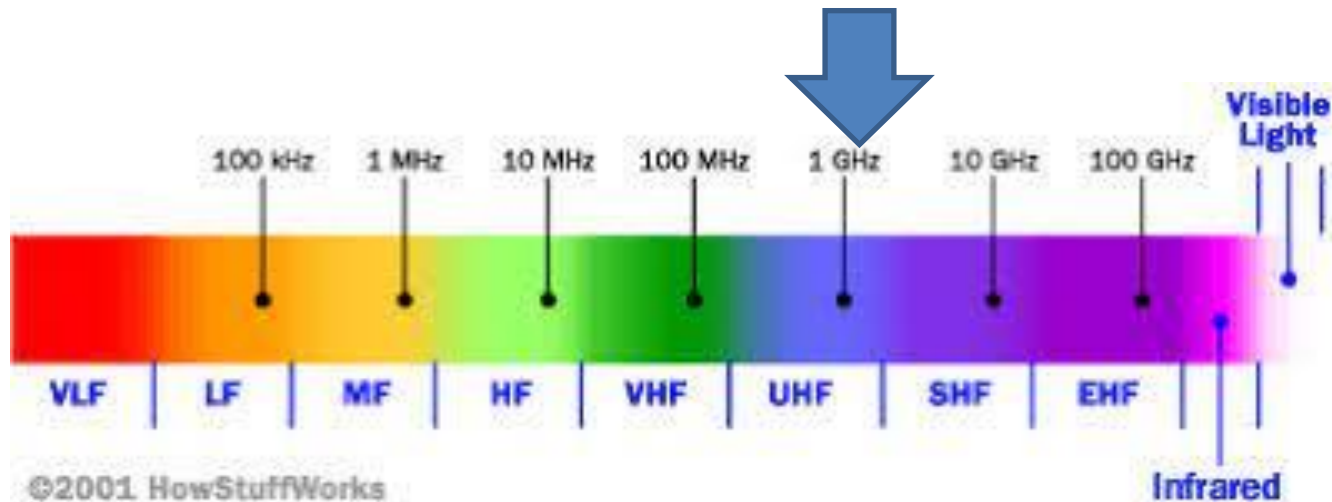
Frequency Range: ~1600 MHz

Primary Use: Battle Tracking and synchronization

Common Lowest Echelon: Light Company/
Mounted Platoon

Modulation: FSK

Max Power: 8W



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TRI-Line Of Sight



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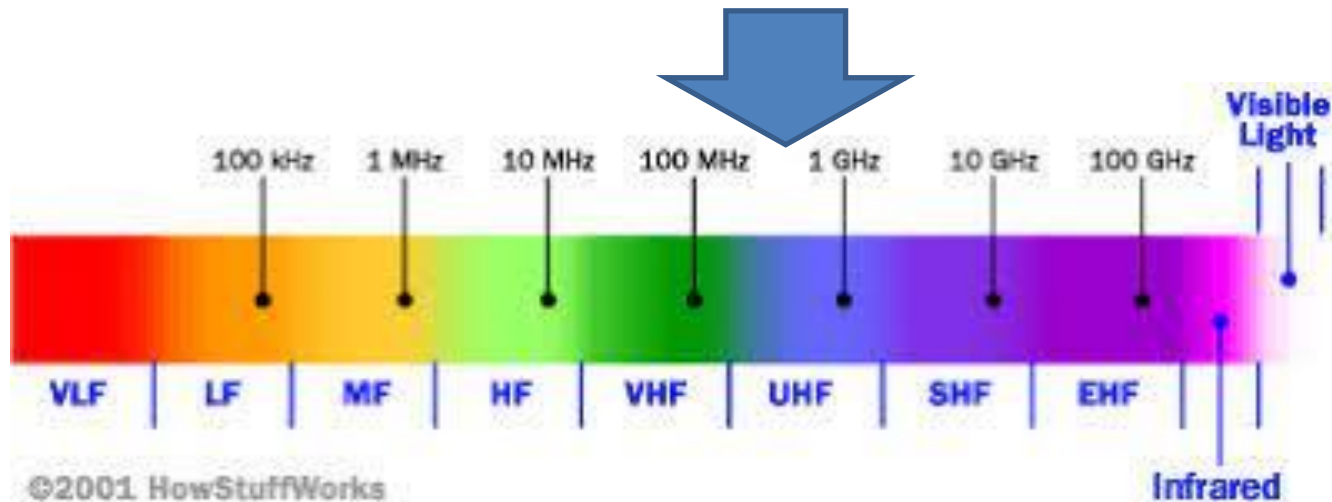
Common Name: TRILOS

Frequency Range: UHF Band

Primary Use: LOS Tactical internet services

Common Lowest Echelon: Battalion

Modulation: FSK, PSK



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Satellite Transport Terminal



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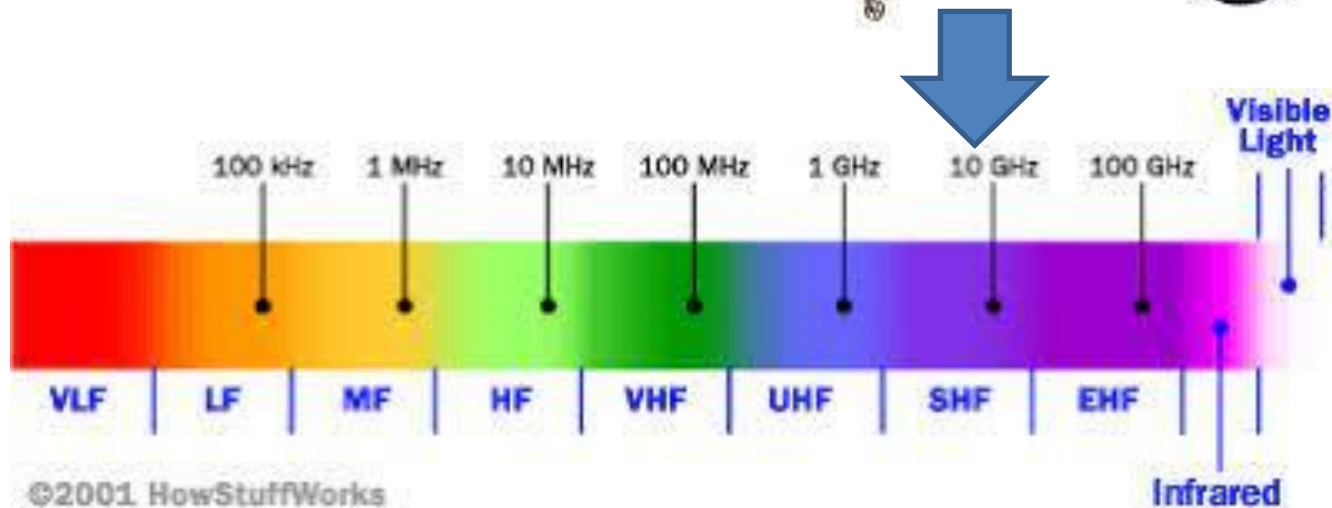
Common Name: STT

Frequency Range: SHF Band

Primary Use: BLOS and Primary transport for Tactical internet services

Common Lowest Echelon: Battalion

Modulation: PSK



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Software Defined Radios (SDR)



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Software Defined Radios (SDR):

- Hardware radio components are implemented through software on a computer or embedded system
- Provides a flexible approach to meet diverse communications needs through software programmable radio technology



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RTL-SDR



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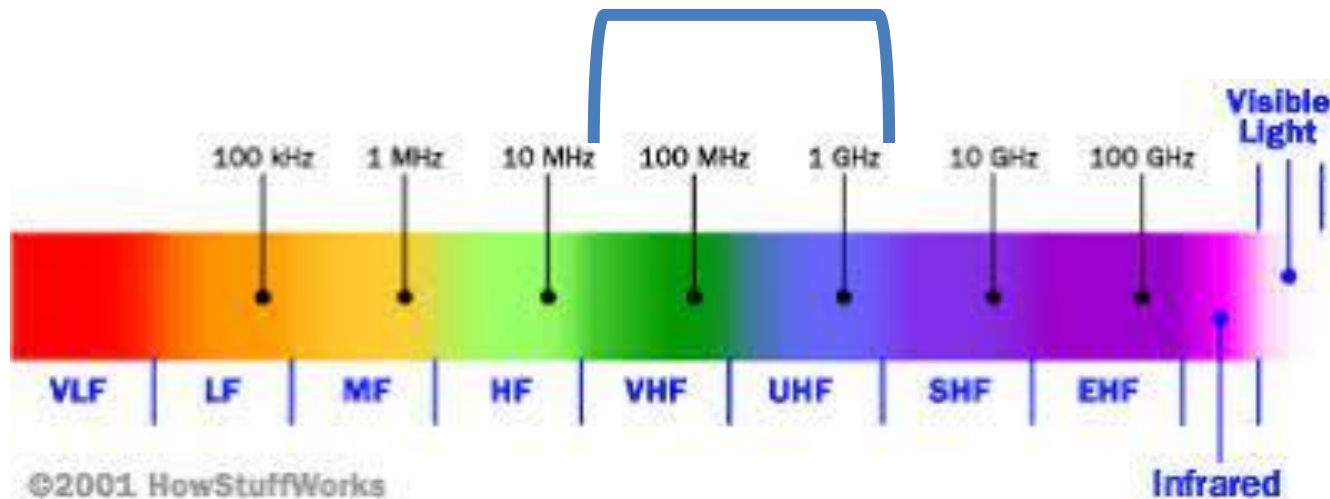
Common Name: RTL-SDR

Frequency Range: 500 kHz – 1.75GHz

Primary Use: Computer-based radio scanner for receiving live radio signals in the area

Applications: Police scanner, EMS monitoring, aircraft/high altitude balloon tracking, digital voice decoding, GPS receiving/decoding

Modulation: AM, FM, SSB, CW



HackRF One SDR



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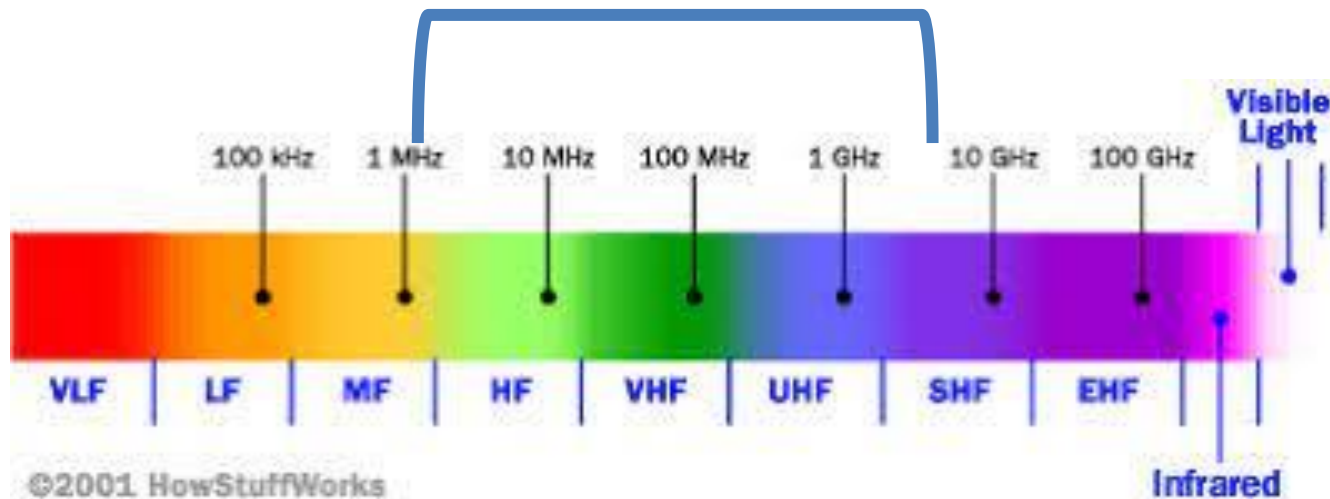
Common Name: HackRF One SDR

Frequency Range: 1 MHz – 6 GHz

Primary Use: SDR primarily used for testing and monitoring RF signals

Applications: Spectrum analyzer, RF research and analysis, security testing, amateur radio

Modes: Transmit, receive, half-duplex



KrakenSDR



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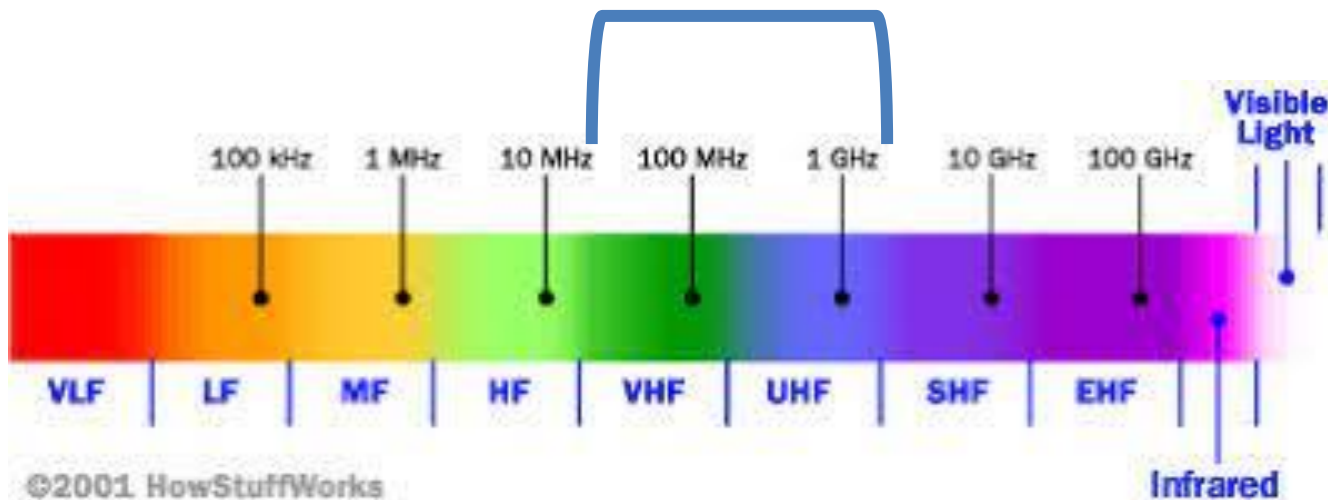
Common Name: KrakenSDR

Frequency Range: 24 MHz – 1.766 GHz

Primary Use: SDR used for direction finding and beamforming

Applications: Tracking assets, locating search and rescue beacons, amateur radio fox hunts

Maximum Channel Bandwidth: 2.56 MHz





Questions?

Summary



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In summary:

- Define the basic components of Radio Communications
- Define Radio Communication Principles
- Define Modulation
- Define Selective Squelch
- Define Basic COMSEC Principles
- Identify Commercial Off the Shelf (COTS) Technologies
- Identify Common Military radios

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References (1 of 2)



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- NAVEDTRA 14189A. *Navy Electricity and Electronics Training Series (NEETS) Module 17: Radio Frequency Communications Principles*. April 2013.
- *Radio Fundamentals Student Guide Volume I*. Fort Sill, OK Electronic Warfare (EW) Command Control Warfare (C2W). June 2010.
- Federal Standard 1037C. *Telecommunications: Glossary of Telecommunications Terms*. Jun 2000.
- MCRP 8-10B.10 (Formerly MCRP 3-40.3B). *Radio Operator's Handbook U.S. Marine Corps*. 2 May 2016.
- MCRP 8-10B.11 (Formerly MCROP 3-40.3C). *Antennal Handbook U.S. Marine Corps*. 2 May 2016.

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References (2 of 2)



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- U.S. Federal Acquisition Regulations.
<https://www.acquisition.gov/content/2101-definitions#i1125359>.
FAC Number/Effective Date: FAC_2019-01/01-22-2019. Far part 2 subpart 2.1 Subtopic 2.101.
- U.S. Federal Communication Commission Wireless Bureau.
<https://www.fcc.gov/wireless/bureau-divisions/mobility-division/family-radio-service-frs> Rule Part 47 C.F. R, Part 95. September 20, 2017.
- U.S. Federal Communication Commission Wireless Bureau.
<https://www.fcc.gov/general-mobile-radio-service-gmrs>. Rule Part 47 C.F.R, part 95 Subpart E FCC Form 605. April 28, 2017.

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